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Element-free numerical manifold method based on the GZZ constitutive model for stability analysis of deeply buried jointed tunnels

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ABSTRACT

To accurately characterize the mechanical response of deeply buried rock masses under complex stress states and the discontinuous effect between joints, the smoothed GZZ strength criterion is introduced into the continuous-discontinuous computational framework of the Element-Free Numerical Manifold Method (E-NMM). The numerical implementation of the GZZ constitutive model is achieved based on the return mapping algorithm, and the relaxation iteration method is adopted to solve the convergence problem of the double nonlinearity problem. Through a systematic analysis of the influences of joint distribution (single joint, double joints), lateral pressure coefficient, and Geological Strength Index (*GSI*) on tunnel stability, it is found that joint distribution plays a dominant role in the morphology of the plastic zone and the characteristics of the displacement field, with an influence weight far exceeding that of the lateral pressure coefficient; an increase in the *GSI* value can significantly improve rock mass integrity, inhibit the expansion of the plastic zone, and reduce surrounding rock deformation.

1. Introduction

The geomechanical environment of deeply buried rock masses exhibits significant complexity and particularity. Compared with shallow engineering, the ground stress field it faces is more intricate, typically characterized by high confining pressure and a triaxial unequal pressure stress state (Zhu et al., 2019; Shi and Zhang, 2023; Wang et al., 2020). In addition to the core characteristic of high ground stress, deeply buried rock masses have undergone a long geological evolution process and generally develop a large number of natural discontinuities such as joints, fractures, and faults. The existence of these discontinuities seriously damages the integrity of rock masses, endowing them with distinct heterogeneity and anisotropy (Zhang et al., 2024; Jiang et al., 2024). Accurately revealing the mechanical behavior characteristics of deeply buried jointed rock masses is of crucial theoretical value and engineering significance for guiding the design optimization and construction of deep engineering (Qian and Zhou, 2018).

The rock mass constitutive model is a core tool for describing the mechanical response of rock masses, and its rationality directly determines the reliability of numerical calculation results (Kalos and Kavvadas, 2017). To accurately characterize the strength properties and deformation behavior of rock masses under complex stress states, scholars at home and abroad have conducted extensive research and proposed a series of rock constitutive models. Early classical strength criteria, such as the Mohr-Coulomb criterion, are based on the single shear plane failure theory, featuring a simple form and easy access to parameters. They have been widely applied in engineering, but it is difficult for this criterion to accurately describe the failure law of rock masses under deep high ground stress conditions (Lu et al., 2010). Later, Hoek and Brown proposed the H-B strength criterion (Hoek and Brown, 1980). This criterion introduces parameters reflecting rock quality characteristics to better describe the strength variation of intact rock under different confining pressures. However, the H-B criterion does not consider the structural characteristics of rock masses. To address this, the generalized H-B (GHB) strength criterion was proposed (Hoek and Brown, 1997). Nevertheless, the GHB criterion fails to fully account for the influence of the intermediate principal stress on rock mass strength. To make up for this defect, Zhang and Zhu put forward the GZZ strength criterion (Zhang and Zhu, 2007; Zhang, 2008). However, the GZZ criterion has non-smooth and non-fully convex mathematical properties (Clausen and Damkilde, 2008). Cai (Cai et al., 2021) et al. proposed the smoothed GZZ criterion, which improves the smoothness

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and convexity of the criterion, laying a foundation for the engineering application of the GZZ criterion. Subsequently, Cai (Cai et al., 2022) et al. analyzed the three-dimensional spatial effect of deeply buried rock tunnels based on the GZZ criterion. Some scholars have introduced the GZZ model into FLAC3D and conducted relevant numerical studies (Chen et al., 2021; Xiao et al., 2024). Wei (Wei et al., 2026) incorporated the GZZ constitutive model into 3DEC and explicitly accounted for the interaction between the GZZ criterion and joints.

Rock mass numerical simulation methods are important means for tunnel stability analysis. Based on the assumption of rock mass continuity, they can be divided into two categories: continuous medium numerical methods and discontinuous medium numerical methods. The core assumption of continuous medium methods is that rock masses are continuous and homogeneous media, and the governing equations are solved based on continuous medium mechanics theory (Zhu et al., 2017; An et al., 2012). However, when there are discontinuities such as joints and fractures in rock masses, continuous medium methods are difficult to accurately simulate the discontinuous deformation behaviors such as sliding and opening of discontinuities. Discontinuous medium methods regard rock masses as a collection of discrete blocks. By simulating the contact and movement between blocks, they can intuitively reflect the influence of discontinuities such as joints on the mechanical behavior of rock masses, but their computational efficiency is relatively low (Xia et al., 2025; Jiang et al., 2025). Among existing numerical methods, Discontinuous Deformation Analysis (DDA) has attained considerable advances in the analysis of jointed tunnels (Jiao et al., 2025; Yang et al., 2023; Fan et al., 2023); nevertheless, this approach is still confined to elastic rock constitutive models.

To overcome the inherent defects of continuous medium methods and discontinuous medium methods, Shi proposed the Numerical Manifold Method (NMM), which can uniformly handle continuous and discontinuous problems through a dual cover system (Shi, 1991). The NMM has found extensive applications in diverse fields (Wu et al., 2022; Wu et al., 2020; Wu et al., 2025; Wu et al., 2020). In the field of tunnel engineering, Wu (Wu and Wong, 2014) introduced a new bolt algorithm into the NMM and carried out corresponding analyses, whereas Xu (Xu et al., 2021; Xu et al., 2022) investigated tunnel stability and surrounding rock reinforcement based on an improved NMM. However, the NMM involves complex geometric calculations in the process of generating manifold elements (Yi et al., 2024; Chen and Li, 2015). For this reason, an Element-Free Numerical Manifold Method (E-NMM) was proposed. By eliminating the specific generation of manifold elements and replacing each manifold element with a point in space, this method effectively avoids complex geometric calculations (Zhang et al., 2026).

Although the GZZ strength criterion has significant advantages in characterizing the mechanical behavior of rock masses under complex stress states, its current application is mainly limited to continuous medium numerical methods such as the finite element method and finite difference method, and has not been introduced into the continuous-discontinuous unified computational framework. Since the stability of deeply buried jointed tunnels is affected by the coupling of continuous deformation of rock masses and discontinuous sliding of joints, traditional continuous medium methods are difficult to accurately simulate this complex process. As an efficient numerical method capable of uniformly handling continuous and discontinuous problems, E-NMM provides an ideal platform for expanding the engineering application of the GZZ criterion. Based on this, this paper introduces the GZZ material model into NMM and applies it to the stability analysis of deeply buried jointed tunnels.

The chapter structure of this paper is arranged as follows: Section 2 briefly reviews the basic principles of the E-NMM; Section 3 elaborates on the numerical implementation process of the GZZ constitutive model; Section 4 verifies the accuracy and reliability of the proposed method; Section 5 establishes a numerical model of deeply buried jointed tunnels and systematically analyzes the stability of jointed tunnels; Section 6 summarizes the main research conclusions of this paper.

2. Brief review of the E-NMM

The NMM consists of two sets of cover systems: the Mathematical Cover (MC) and the Physical Cover (PC). The MC is a set of arbitrarily defined shape patches, as long as they can completely cover the entire computational domain. These patches are called Mathematical Patches (MPs), and weight functions are defined on the MPs, with the center of each MP being a star point. The PC is the intersection of the MC and the computational domain, and Physical Patches (PPs) are generated by subdividing the MPs. When different PPs overlap with each other, Manifold Elements (MEs) are formed.

The basic computational unit of the NMM is the ME. The main goal of preprocessing is to obtain all MEs and the degrees of freedom corresponding to each ME (i.e., the star points controlling each ME), as shown in Fig. 1. For each complete MC element, the shape of its ME is exactly the same as the element formed by the star points (i.e., the FE element). At this time, the calculation process of this ME is completely equivalent to that of the Finite Element (FE) method. For a MC element that is cut, it will generate two MEs, and different star points are assigned to the two MEs respectively. However, each ME is still controlled by the star points forming a complete FE element shape. At this time, the calculation process for these two MEs is still equivalent to the FE method because it is still composed of a standard FE element. The only difference is that the integration domain (ME) has changed. Therefore, in the NMM, the element stiffness matrix is calculated as (for first-order linear elements):

$$k_{ME} = B^T D B \bullet S_{ME} \tag{1}$$

where B is the element strain matrix, D is the element material matrix, and S_{ME} is the area/volume value (integral value) of the ME controlled by each standard FE element.

It can be seen from the above that in the NMM, the main function of the ME is to provide the integration domain during the calculation of the element stiffness matrix each time. Therefore, as long as the integral value can be obtained, specific ME elements do not need to be generated in the specific preprocessing and calculation process of the NMM, so no complex geometric calculations will be involved. This method is called the Element-Free Numerical Manifold Method (E-NMM).

Compared with the NMM (Fig. 1), the E-NMM can be summarized as shown in Fig. 2. The E-NMM only eliminates the generation of MEs, thereby avoiding complex geometric calculations, and replaces each ME with a point in space (Numerical Manifold Point, NMP). That is to say, the calculation process of the E-NMM is exactly the same as that of the NMM, and Eq. (1) is still used for calculation. The only difference is that SME in the equation becomes the result pre-calculated through the voxel grid. For the specific calculation process, refer to Refs. (Zhang et al., 2026; Zhang et al., 2026).

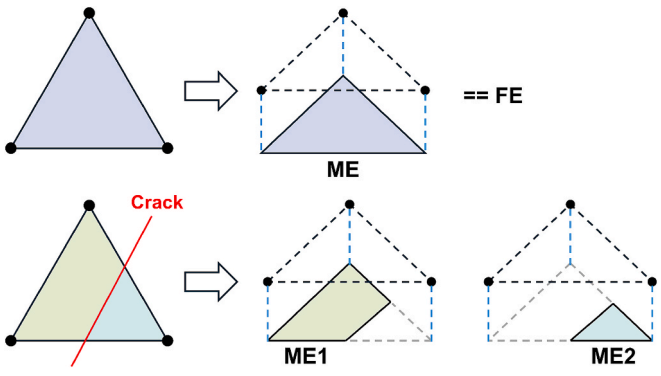


Fig. 1. Relationship between MEs and star points in the NMM.

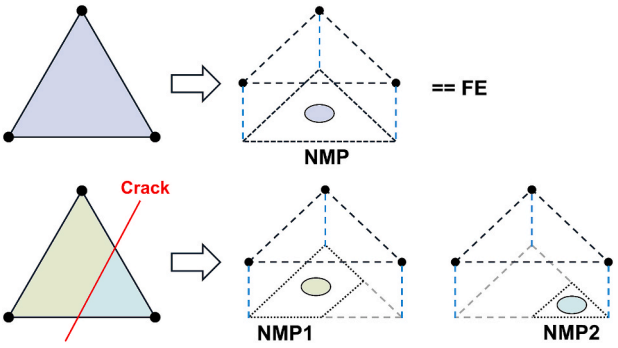


Fig. 2. Relationship between NMPs and star points in the E-NMM.

3. Numerical implementation of the GZZ constitutive model

3.1. GZZ strength criterion

The GZZ criterion was proposed based on the GHB criterion (Clausen and Damkilde, 2008). Hoek and Brown (Hoek and Brown, 1980) put forward the H-B strength criterion in 1980, which was initially applied to describe the strength variation of intact rock under different confining pressures. Its expression is as follows:

$$\sigma_1 = \sigma_3 + \sigma_c\left(m_i\frac{\sigma_3}{\sigma_c} + 1\right)^{0.5} \tag{2}$$

where: σ_1 and σ_3 are the maximum and minimum principal stresses, respectively; σ_c is the uniaxial compressive strength of rock; m_i is a material parameter reflecting rock quality characteristics.

Furthermore, the Geological Strength Index (*GSI*) describing the structural characteristics of rock masses is introduced into the H-B criterion, and the generalized H-B (GHB) strength criterion is proposed (Hoek and Brown, 1997). The expression of the GHB strength criterion is as follows:

$$\sigma_1 = \sigma_3 + \sigma_c\left(m_b\frac{\sigma_3}{\sigma_c} + s\right)^{a} \tag{3}$$

$$m_b = m_i e^{\frac{GSI-100}{28-14D}} \tag{4}$$

$$s = e^{\frac{GSI-100}{9-3D}} \tag{5}$$

$$a = 0.5 + \frac{e^{-\frac{GSI}{15}} - e^{-\frac{20}{3}}}{6} \tag{6}$$

where: m_b, s, and a are empirical parameters reflecting lithological and rock mass structural characteristics; D is the blasting disturbance coefficient. *GSI* is determined by the rock mass structure grade and surface conditions.

However, the GHB criterion ignores the influence of the intermediate principal stress σ_2 on rock mass strength. To address this, Zhang and Zhu (Zhang and Zhu, 2007) proposed the Zhang-Zhu strength criterion, whose expression is as follows:

$$\frac{9}{2\sigma_c}\tau_{oct}^2 + \frac{3}{2\sqrt{2}}m_b\tau_{oct} - m_b\sigma_{m,2} = s\sigma_c \tag{7}$$

$$\tau_{oct} = \frac{1}{3}\sqrt{(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2} \tag{8}$$

$$\sigma_{m,2} = \frac{\sigma_1+\sigma_3}{2} \tag{9}$$

where: τ_{oct} is the octahedral shear stress; $\sigma_{m,2}$ is the effective mean normal stress.

Further, Zhang (Zhang, 2008) improved the Zhang-Zhu criterion and proposed the GZZ criterion:

$$\frac{1}{\sigma_c^{\frac{1}{a}-1}}\left(\frac{3}{\sqrt{2}}\tau_{oct}\right)^{\frac{1}{a}} + \frac{m_b}{2}\left(\frac{3}{\sqrt{2}}\tau_{oct}\right) - m_b\sigma_{m,2} - s\sigma_c = 0 \tag{10}$$

The parameter values in the formula are consistent with those in the GHB criterion.

However, the GZZ criterion has non-smooth and non-fully convex properties, making it difficult to converge numerically. For this reason, Cai et al. (Cai et al., 2021) proposed the smoothed GZZ criterion:

$$\varphi(\sigma) = \frac{1}{\sigma_c^{\frac{1}{a}-1}}\left(\frac{3}{\sqrt{2}}\tau_{oct}\right)^{\frac{1}{a}} + \frac{m_b}{2}\left(\frac{3}{\sqrt{2}}\tau_{oct}\right) - m_b\left(\sigma_{m,2}\right)_{new-GZZ} - s\sigma_c = 0 \tag{11}$$

$$\left(\sigma_{m,2}\right)_{new-GZZ} = \frac{1}{2}\left(I_1 - \frac{I_1I_2 - 9I_3}{6J_2}\right) \tag{12}$$

where: I_1, I_2, and I_3 are the first, second, and third stress invariants, respectively; J_2 is the second deviatoric stress invariant.

Cai et al. (Cai et al., 2021) have given the mathematical expression of the plastic potential function for the smoothed GZZ criterion:

$$g = \frac{1}{\sigma_c^{\frac{1}{a}-1}}\left(\sqrt{3J_2}\right)^{\frac{1}{a}} + \left(\frac{3+sin3\theta_\sigma}{2\sqrt{3}}\right)m_b\sqrt{J_2} - \frac{m_bI_1}{3} - s\sigma_c = 0 \tag{13}$$

$$\theta_\sigma = tan^{-1}\left[\frac{\sigma_1+\sigma_3-2\sigma_2}{\sqrt{3}\left(\sigma_1-\sigma_3\right)}\right] \tag{14}$$

where: θ_σ is the stress Lode angle, expressed by stress invariants as:

$$sin(3\theta_\sigma) = -\frac{3\sqrt{3}}{2J_2^{\frac{3}{2}}}\left(I_3 + \frac{1}{3}I_1J_2 + \frac{1}{27}I_1^3\right) \tag{15}$$

The smoothed GZZ criterion has better smoothness and numerical convergence. In this paper, the material model in E-NMM is developed based on the smoothed GZZ criterion.

3.2. Numerical implementation

Based on the return mapping algorithm commonly used in elasto-plastic material model calculations, the stress and stiffness matrix of each incremental step are updated in the general stress space, thereby realizing the application of the GZZ material model in E-NMM.

First, an elastic trial calculation is performed. Assuming that the strain increment $\Delta\varepsilon$ generated in the incremental step is all elastic strain, the trial stress σ^{tr} is obtained:

$$\sigma^{tr} = \sigma^n + D^e\Delta\varepsilon \tag{16}$$

where: D^e is the elastic matrix; σ^n is the stress at the n-th incremental step.

The yield function is defined according to Eq. (11):

$$f(\sigma) = \varphi(\sigma) \tag{17}$$

If $f(\sigma^{tr}) \le 0$, the generated strain increment is entirely elastic. If $f(\sigma^{tr}) > 0$, plastic deformation occurs in this incremental step, and the trial stress needs to be updated. Due to the existence of the term $(I_1I_2 - 9I_3)/6J_2$ in the smoothed GZZ criterion formula, numerical singularity may occur when the material is in a hydrostatic pressure state. Therefore, in addition to $f(\sigma^{tr}) > 0$, it is also necessary to satisfy $J_2 >$ TOL (TOL is set according to calculation requirements, This paper takes 1×10^{-8}) before performing the plastic stress update calculation.

The plastic strain residual is:

$$R\left(\sigma^{n+1}, \Delta\lambda_{n+1}\right) = -\varepsilon_{n+1}^{p} + \varepsilon_{n}^{p} + \frac{\Delta\lambda_{n+1}(\partial g)}{\partial\sigma^{n+1}} \tag{18}$$

Thus, it is necessary to solve for σ and $\Delta\lambda$ (consistency parameter) through iteration to make $f(\sigma) = 0$ and $R(\sigma, \Delta\lambda) = 0$ until the target converges to the allowable error.

The detailed solving procedure is described below, where the incremental step subscript $n+1$ is omitted:

$$R^{(k)} + \left(\frac{\partial R}{\partial\sigma}\right)^{(k)}\Delta\sigma + \left(\frac{\partial R}{\partial\Delta\lambda}\right)^{(k)}\Delta(\Delta\lambda) = 0\#(19)$$

$$f^{(k)} + \left[\left(\frac{\partial f}{\partial\sigma}\right)^{(k)}\right]^{\mathrm{T}}\Delta\sigma = 0\#(20)$$

$$\sigma^{(k+1)} = \sigma^{(k)} + \Delta\sigma\#(21)$$

$$\Delta\lambda^{(k+1)} = \Delta\lambda^{(k)} + \Delta(\Delta\lambda)\#(22)$$

where k is the current iteration count, $\Delta\sigma$ denotes the stress increment, and $\Delta(\Delta\lambda)$ is the increment of the consistency parameter.

Eq. (19) can be rearranged as:

$$R^{(k)} + [D^{\mathrm{e}}]^{-1}\Delta\sigma + \Delta\lambda^{(k)}\left(\frac{\partial^2 g}{\partial\sigma\partial\sigma}\right)^{(k)}\Delta\sigma + \Delta(\Delta\lambda)\left(\frac{\partial g}{\partial\sigma}\right)^{(k)} = 0\#(23)$$

Rearranging Eq. (23) yields:

$$\Delta\sigma = -A^{(k)}R^{(k)} - \Delta(\Delta\lambda)A^{(k)}\left(\frac{\partial g}{\partial\sigma}\right)^{(k)}\#(24)$$

$$A^{(k)} = \left[[I] + \Delta\lambda^{(k)}[D^{\mathrm{e}}]\left(\frac{\partial^2 g}{\partial\sigma\partial\sigma}\right)^{(k)}\right]^{-1}[D^{\mathrm{e}}]\#(25)$$

where $[I]$ represents the identity matrix. Substituting Eq. (24) into Eq. (20) gives:

$$\Delta(\Delta\lambda) = \frac{f^{(k)} - \left[\left(\frac{\partial f}{\partial\sigma}\right)^{(k)}\right]^{\mathrm{T}}A^{(k)}R^{(k)}}{\left[\left(\frac{\partial f}{\partial\sigma}\right)^{(k)}\right]^{\mathrm{T}}A^{(k)}\left(\frac{\partial g}{\partial\sigma}\right)^{(k)}}\#(26)$$

Substituting Eq. (26) back into Eq. (24), the stress increment $\Delta\sigma$ and consistency parameter increment $\Delta(\Delta\lambda)$ can be fully expressed by variables from the previous iteration step. Afterwards, the stress tensor, consistency parameter and plastic strain are updated as:

$$\begin{cases} \sigma^{(k+1)} = \sigma^{(k)} + \Delta\sigma \\ \Delta\lambda^{(k+1)} = \Delta\lambda^{(k)} + \Delta(\Delta\lambda) \\ \varepsilon^{\mathrm{p}(k+1)} = \varepsilon^{\mathrm{p}(k)} - [D^{\mathrm{e}}]^{-1}\Delta\sigma \end{cases}\#(27)$$

The updated parameters are substituted into the yield function $f^{(k+1)}$ and plastic residual $R^{(k+1)}$ to examine the convergence criterion. If the convergence requirement is satisfied, $\sigma^{(k+1)}$ is determined as the converged stress state for the current incremental step; otherwise, the above iterative procedure is repeated until the objective function converges.

Updating the stiffness matrix during plastic calculation helps accelerate the convergence rate and improve the numerical convergence of the model.

For elastic material states, the consistent elastoplastic constitutive matrix D^{ep} derived from the GZZ criterion is identical to the elastic stiffness matrix D^{e}. The consistent constitutive matrix under plastic conditions is deduced from the incremental forms of stress tensor, strain tensor and plastic strain tensor:

$$\mathrm{d}\sigma = D^{\mathrm{e}}(\mathrm{d}\varepsilon - \mathrm{d}\varepsilon^{\mathrm{p}})\#(28)$$

Based on the plastic flow vector and consistency parameter, the incremental plastic strain tensor is written as:

$$\mathrm{d}\varepsilon^{\mathrm{p}} = \mathrm{d}(\Delta\lambda)\frac{\partial g}{\partial\sigma} + \Delta\lambda\frac{\partial^2 g}{\partial\sigma\partial\sigma}\mathrm{d}\sigma\#(29)$$

Differentiating the yield function $f(\sigma)$ obtains:

$$\mathrm{d}[f(\sigma)] = \left[\frac{\partial f}{\partial\sigma}\right]^{\mathrm{T}}\mathrm{d}\sigma\#(30)$$

Substitution of Eq. (29) into Eq. (28) leads to:

$$\mathrm{d}\sigma = \overline{A}\mathrm{d}\varepsilon - \mathrm{d}(\Delta\lambda)\overline{A}\frac{\partial g}{\partial\sigma}\#(31)$$

$$\overline{A} = \left(I + D^{\mathrm{e}}\Delta\lambda\frac{\partial^2 g}{\partial\sigma\partial\sigma}\right)^{-1}D^{\mathrm{e}}\#(32)$$

Inserting Eq. (31) into the differentiated yield function Eq. (30) yields:

$$\mathrm{d}(\Delta\lambda) = \frac{\left[\frac{\partial f}{\partial\sigma}\right]^{\mathrm{T}}\overline{A}\mathrm{d}\varepsilon}{\left[\frac{\partial f}{\partial\sigma}\right]^{\mathrm{T}}\overline{A}\frac{\partial g}{\partial\sigma}}\#(33)$$

By substituting Eq. (33) into Eq. (29) and further into Eq. (28), the final expression of consistent elastoplastic constitutive matrix is acquired:

$$D^{\mathrm{ep}} = \overline{A} - \frac{\overline{A}\frac{\partial g}{\partial\sigma}\left[\frac{\partial f}{\partial\sigma}\right]^{\mathrm{T}}\overline{A}}{\left[\frac{\partial f}{\partial\sigma}\right]^{\mathrm{T}}\overline{A}\frac{\partial g}{\partial\sigma}}\#(34)$$

It is worth noting that E-NMM is a unified continuous-discontinuous calculation method. Therefore, contact and material nonlinearity need to be considered uniformly in the calculation. The joint element model is adopted for contact characterization, with its constitutive matrix expressed as:

$$D_j = \begin{bmatrix} K_n & 0 \\ 0 & K_s \end{bmatrix}\#(35)$$

where K_n and K_s denote the normal stiffness and shear stiffness of the joint element, respectively.

This paper mainly focuses on statics, and the main flow of E-NMM is shown in Fig. 3.

Due to contact nonlinearity and material nonlinearity, the standard Newton-Raphson iteration often diverges due to sudden stiffness changes and state transitions. This paper adopts the relaxation iteration method, which controls the displacement increment by introducing a factor η ($0 < \eta \leq 1$) ($U_{k+1} = U_k + \eta \times \Delta U_k$). This method effectively suppresses iterative oscillation, expands the convergence domain, and significantly improves solution stability. Although it sacrifices part of the convergence speed, it can reliably capture the complex nonlinear response process.

4. Validation of the research methodology

Two numerical examples are employed to validate the contact nonlinearity and material nonlinearity in the E-NMM, respectively.

4.1. Validation of joint element contact

A cantilever beam composed of two identical stacked beams is modeled. Both beams share material parameters: elastic modulus $E =$

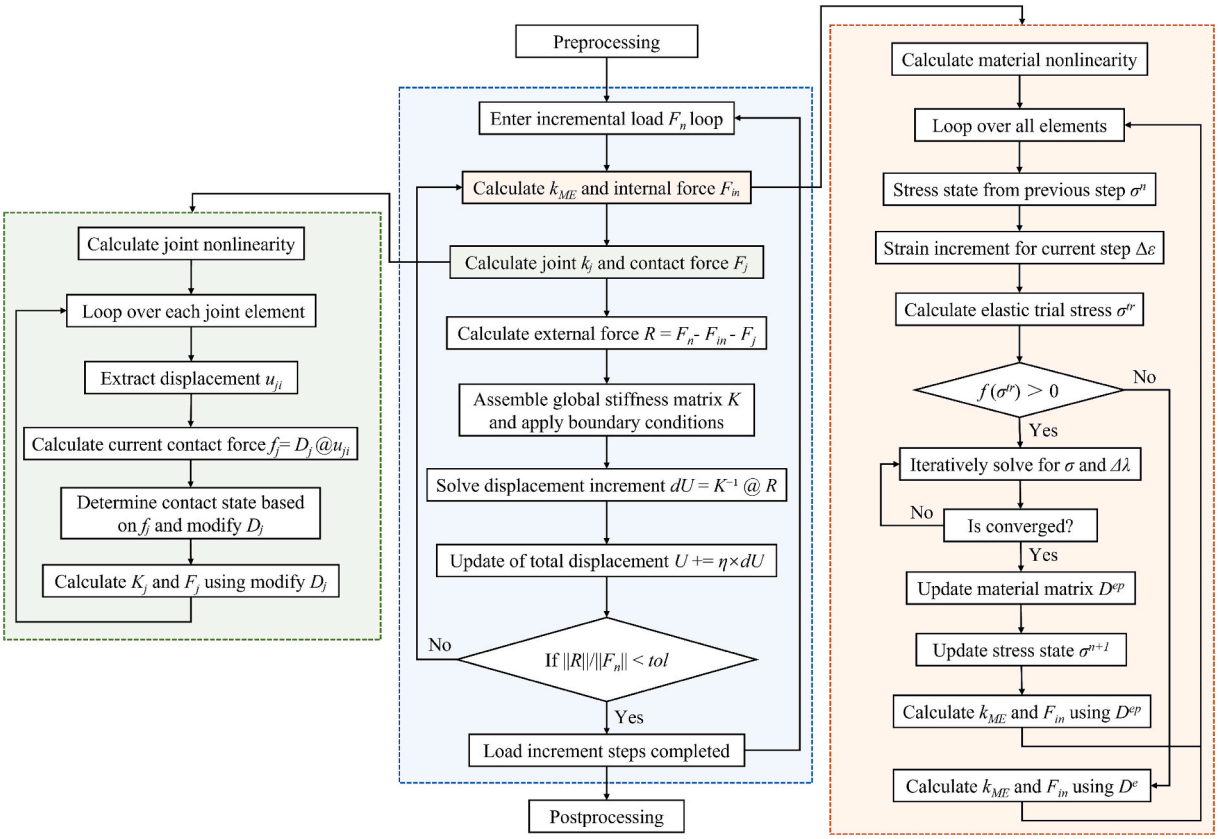


Fig. 3. Flowchart of E-NMM calculation.

1500 MPa, Poisson's ratio $\nu = 0.25$, and are subjected to a concentrated force of 1500 N. Following the setup in Ref. (Sun et al., 2023), upward vertical force is considered. The model dimensions are illustrated in Fig. 4. A total of 810 NMPs were generated in this case. The monitoring point is located at the middle position of the stacked beam (10, 1).

The normal gap distance between the two beams under vertical concentrated forces is analyzed in Fig. 5. The results demonstrate high computational accuracy of the proposed method, aligning closely with Ref. (Sun et al., 2023), thereby validating its effectiveness.

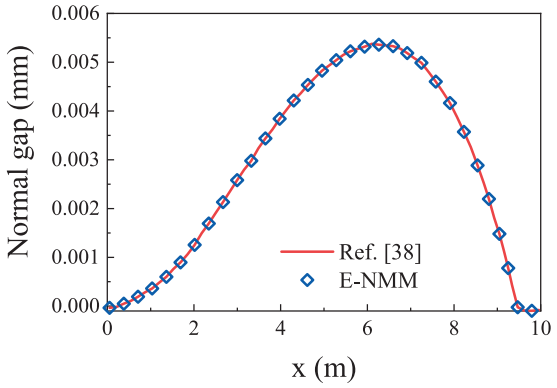


Fig. 5. Normal gap between contact interface applied an upward load.

4.2. Validation of the GZZ constitutive model

A tunnel excavation model is established using a quarter-circle geometry with a model radius 120 m and a tunnel excavation radius of 8 m. The model dimensions and boundary conditions are shown in Fig. 6. A total of 7017 NMPs were generated in this case. Monitoring points are arranged radially. A hydrostatic pressure of 30 MPa is applied at the outer boundary, and a support pressure of 2 MPa is applied on the tunnel surface. The rock mass properties are as follows: Elastic modulus $E = 5$ GPa, Poisson's ratio $\nu = 0.35$, $m_i = 8$, $D = 0$, $GSI = 80$, $\sigma_c = 40$ MPa.

The stress distribution obtained from the numerical simulation is compared with the semi-analytical solution proposed by Su et al. (Su et al., 2020), as shown in Fig. 7. The numerical results show good agreement with the analytical solution, demonstrating the high accuracy of the numerical computations presented in this study.

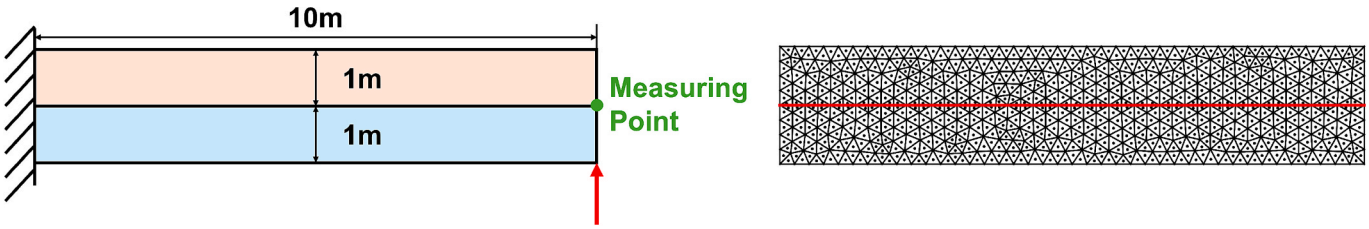


Fig. 4. Stacked beams computational model.

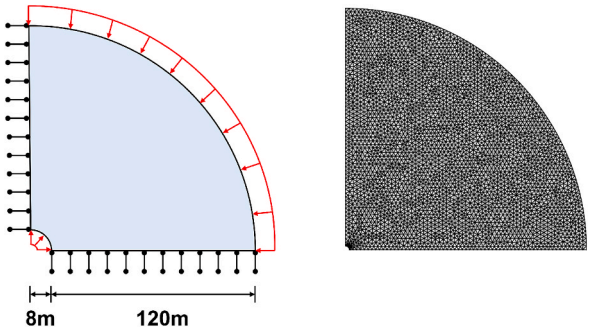


Fig. 6. Computational model for tunnel excavation.

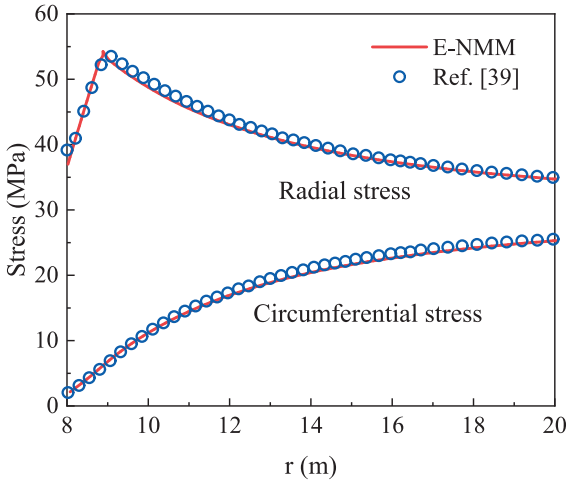


Fig. 7. Comparison between numerical and analytical solutions.

5. Application and analysis

5.1. Numerical model of deeply buried jointed tunnels

A calculation model was established based on the geometric dimensions and similarity ratio-scaled parameters of the physical model test in Ref. (Cai et al., 2023). The relatively low values of rock mass parameters and in-situ stress are inherent characteristics of laboratory physical model tests, where low-strength similar materials and scaled stress fields are adopted. The rock mass parameters are as follows: Elastic modulus $E = 0.09$ GPa, Poisson's ratio $\nu = 0.32$, $m_i = 21$, $D = 0$, $GSI = 100$, and uniaxial compressive strength $\sigma_c = 0.016$ MPa. As reported in Ref. (Cai et al., 2022), the GZZ criterion yields significantly better calculation results for deep-buried conditions compared to other constitutive models. Therefore, only deep-buried scenarios are considered in this study. Load boundaries were applied to the top and both sides of the model, with two lateral pressure coefficients $k = \sigma_H / \sigma_V$, where $\sigma_V = 0.855$ MPa, $k = 0.5$ and $k = 1.5$, and fixed displacement constraints in the vertical direction were imposed at the bottom. A two-dimensional plane strain model was adopted, and the model dimensions and boundary conditions are shown in Fig. 8.

Numerical simulations in Ref. (Cai et al., 2023) were conducted using the finite element method without considering the influence of different joint distributions on tunnel stability. In contrast, the E-NMM adopted in this paper can uniformly handle continuous and discontinuous problems, enabling the introduction of joint distribution into the calculation model. The joint normal stiffness was set to 0.9 GPa and shear stiffness as 0.09 MPa. After parameter tuning, the relaxation iteration factor $\eta = 0.7$ is adopted for the coupled calculation of contact and material nonlinearity to balance numerical convergence and computational efficiency.

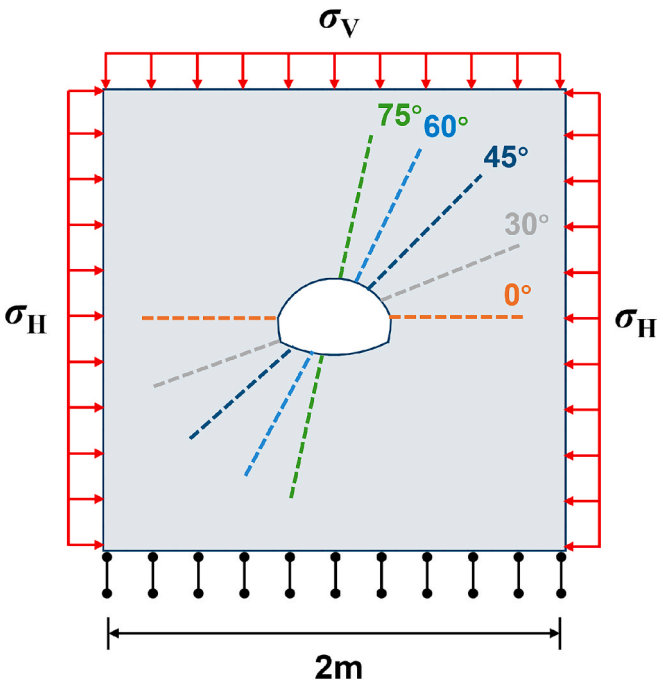


Fig. 8. Calculation model of jointed tunnels.

Single-joint distributions with dip angles of 0°, 30°, 45°, 60°, and 75°, as well as double-joint distributions combining 0° with other angles, were analyzed. Finally, the influence of *GSI* on tunnel behavior in the presence of joints was investigated by varying the *GSI* value.

It is noteworthy that a two-dimensional plane strain model is employed in this work. Although plane strain problems feature a two-dimensional geometric configuration, they physically belong to a special category of three-dimensional stress conditions. The stress tensor at each integration point includes four non-zero components $(\sigma_x, \sigma_y, \sigma_z, \tau_{xy})$. Rather than being preset to zero or a fixed value, the out-of-plane normal stress σ_z is defined as an unknown variable and solved iteratively under the out-of-plane strain constraint $\Delta\varepsilon_z = 0$ during constitutive integration. The trial stress at each calculation step is expressed as:

$$\sigma_z^{trial} = \sigma_z^{old} + \lambda(\Delta\varepsilon_x + \Delta\varepsilon_y) \#(36)$$

where λ is the Lamé constant. Consequently, the three principal stresses $(\sigma_1, \sigma_2, \sigma_3)$ can be rigorously calculated, providing complete stress information for invoking the GZZ strength criterion.

To validate the correctness and advantages of the adopted 2D plane strain model, a 3D counterpart with identical geometric size and material parameters is built for numerical computation, where the lateral pressure coefficient is specified as $k = 1.0$. The obtained vertical displacement nephograms are plotted in Fig. 9, which reveals nearly identical calculation results between the 2D and 3D models.

Crown and invert displacements of the tunnel are extracted and compared with the test data from Ref. (Cai et al., 2023), as summarized in Table. 1. A standard elastic model is also included for comparison to highlight the necessity of introducing the GZZ criterion. The calculated displacements from the GZZ model show good consistency between 2D and 3D simulations and fit well with experimental measurements, while the elastic model produces much smaller displacement values and cannot reflect the plastic deformation around the tunnel.

Meanwhile, the computational time of the 2D and 3D models is recorded. All numerical runs are implemented on an Intel Core i7-13700KF CPU with 32 GB memory. The 3D model consists of 22,687 meshes with a total runtime of 483 s, whereas the 2D model only

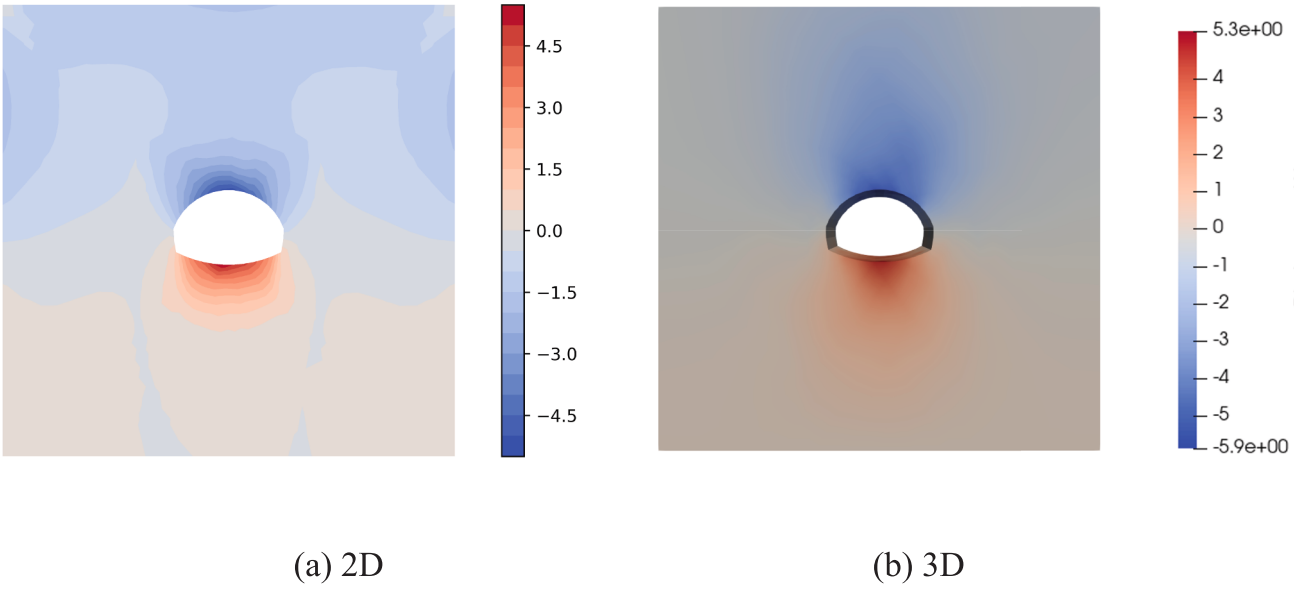


Fig. 9. Vertical displacement nephograms of 2D and 3D models.

Table 1
Comparison of tunnel displacements between different models and physical test.

	Physical result (mm) (Cai et al., 2023)	GZZ-3D (mm)	GZZ-2D (mm)	Elastic (mm)
Crown	46.1	51.0	52.1	19.8
Invert	–	50.3	51.2	4.8

contains 1494 meshes and takes merely 32 s. The GZZ constitutive model incorporates the complicated return mapping algorithm and hence suffers from low computational efficiency, especially for 3D models with massive degrees of freedom. The dense mesh layout of the 3D model is inevitable; accordingly, the 2D model can drastically accelerate calculation while ensuring acceptable accuracy when the longitudinal variation of the tunnel is not considered.

5.2. Results and discussion

First, the calculation results for the case without joints were compared with those in Ref. (Cai et al., 2023), and the plastic zone distribution was extracted as shown in Fig. 10. It can be observed that the plastic zone distribution around the tunnel calculated in this paper is basically consistent with that in the reference under different lateral pressure coefficients, demonstrating the reliability of the proposed method. When the lateral pressure coefficient is small, the plastic zone is concentrated on both sides of the tunnel; as the lateral pressure coefficient increases, the plastic zone shifts to the upper and lower parts of the tunnel.

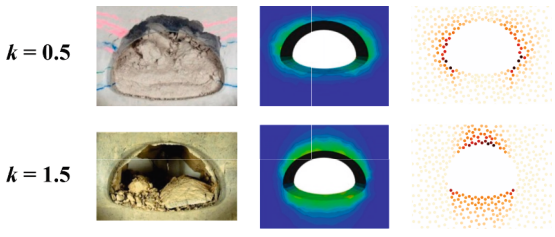


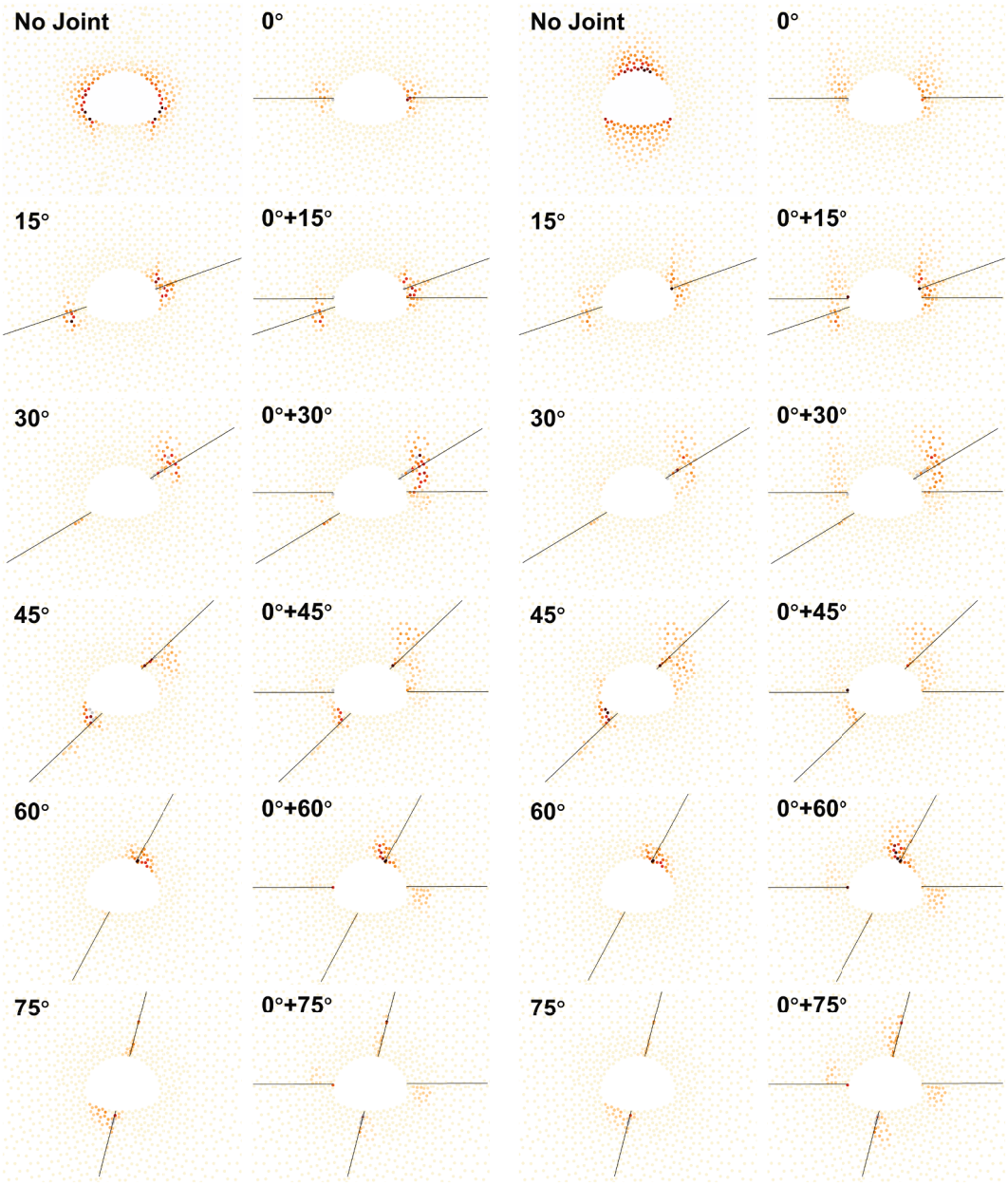
Fig. 10. Comparison of plastic zone distribution without joints.

Furthermore, the plastic zone distribution around the tunnel under different lateral pressure coefficients and joint distributions was extracted, as shown in Fig. 11, and the vertical displacement nephograms are presented in Fig. 12.

In the absence of joints, the plastic zone distribution around the tunnel is mainly controlled by the deep ground stress, i.e., different lateral pressure coefficients lead to different plastic development. However, a horizontal comparison of the plastic zone distributions under the two k values reveals that although the plastic zone is slightly larger when $k = 1.5$ than when $k = 0.5$, the plastic zone development is almost identical in the presence of joints. Therefore, it can be concluded that the influence of joints on the stress distribution around the tunnel is far greater than that of the ground stress level. This is because the existence of joints destroys the continuity of the rock mass, resulting in a fundamental change in the stress transfer path, and the joint surfaces become the weak planes prone to failure. At this time, even under different ground stress conditions, the shear failure mode of the joint surfaces still dominates, leading to convergent plastic zone development.

Comparing different joint distributions, it is found that the plastic zone rotates with the rotation of the joint dip angle. In the presence of joints, the plastic zone extends along the joint direction, forming an asymmetric and directionally expanding failure pattern. When two sets of joints overlap, the scope of the plastic zone further increases. When the angle between the two sets of joints is small, the plastic zone is mostly concentrated near the angle; at this time, the rock bridge in this area bears higher shear stress, leading to concentrated development of the plastic zone near the angle. As the angle gradually increases, the rock mass tends to choose the failure path with the minimum energy consumption, and the controlling effect of a single set of joints reappears, with the plastic zone distribution returning to the single-joint case.

The distribution law of vertical displacement around the tunnel is highly consistent with the plastic zone morphology. This is because the plastic zone, as the core area of stress release and deformation concentration, directly determines the distribution characteristics of the displacement field. In the presence of joints, the continuity of the rock mass is destroyed, resulting in a significant asymmetric displacement pattern around the tunnel. It is worth noting that although an increase in the lateral pressure coefficient intensifies the overall displacement amplitude by increasing the horizontal stress level, the spatial distribution pattern of displacement is mainly controlled by the geometric



(a) $k = 0.5$ (b) $k = 1.5$

Fig. 11. Plastic zone distribution around the tunnel under different lateral pressure coefficients and joint distributions.

characteristics of the joints rather than the ground stress anisotropy. In particular, the displacement field shows an obvious unidirectional deflection under a single set of joints, while the overlapping of two sets of joints forms a complex block system due to mutual cutting between joints, leading to a more irregular displacement distribution concentrated in the joint intersection area. This again confirms that in jointed rock masses, the spatial distribution of structural planes exerts a controlling effect on tunnel deformation that far exceeds the anisotropy of the original rock stress field. The deformation mechanism transforms from overall elastoplasticity to shear sliding and opening of the dominant structural planes, thus manifesting the dominant influence of joints on tunnel stability at the engineering scale.

As shown in Figs. 13 and 14, As the lateral pressure coefficient k increases from 0.5 to 1.5, the absolute value of the tunnel's vertical displacement rises significantly. When horizontal stress is much higher than vertical stress, the surrounding rock around the tunnel suffers intensified compressive squeezing and develops severe convergent deformation. Meanwhile, joints remarkably amplify such deformation responses. Displacement under the joint-free condition is substantially smaller than that with joints, and the displacement fluctuates nonlinearly against joint dip angles, showing evident anisotropy.

However, regardless of the k value, the core characteristics of displacement distribution are controlled by joints: under a single set of joints, the displacement peak is concentrated near a dip angle of 30° and 60°, indicating that this angle is most prone to shear sliding and stress concentration along the joint surface; when two sets of joints overlap results in more irregular displacement distribution, and the maximum crown settlement is slightly larger than that for a single joint set, which is attributed to the discrete block system formed by the cutting of multiple sets of joints, intensifying local deformation. It is worth noting that the asymmetry between crown and invert displacements is particularly significant in the presence of joints, which is highly sensitive to

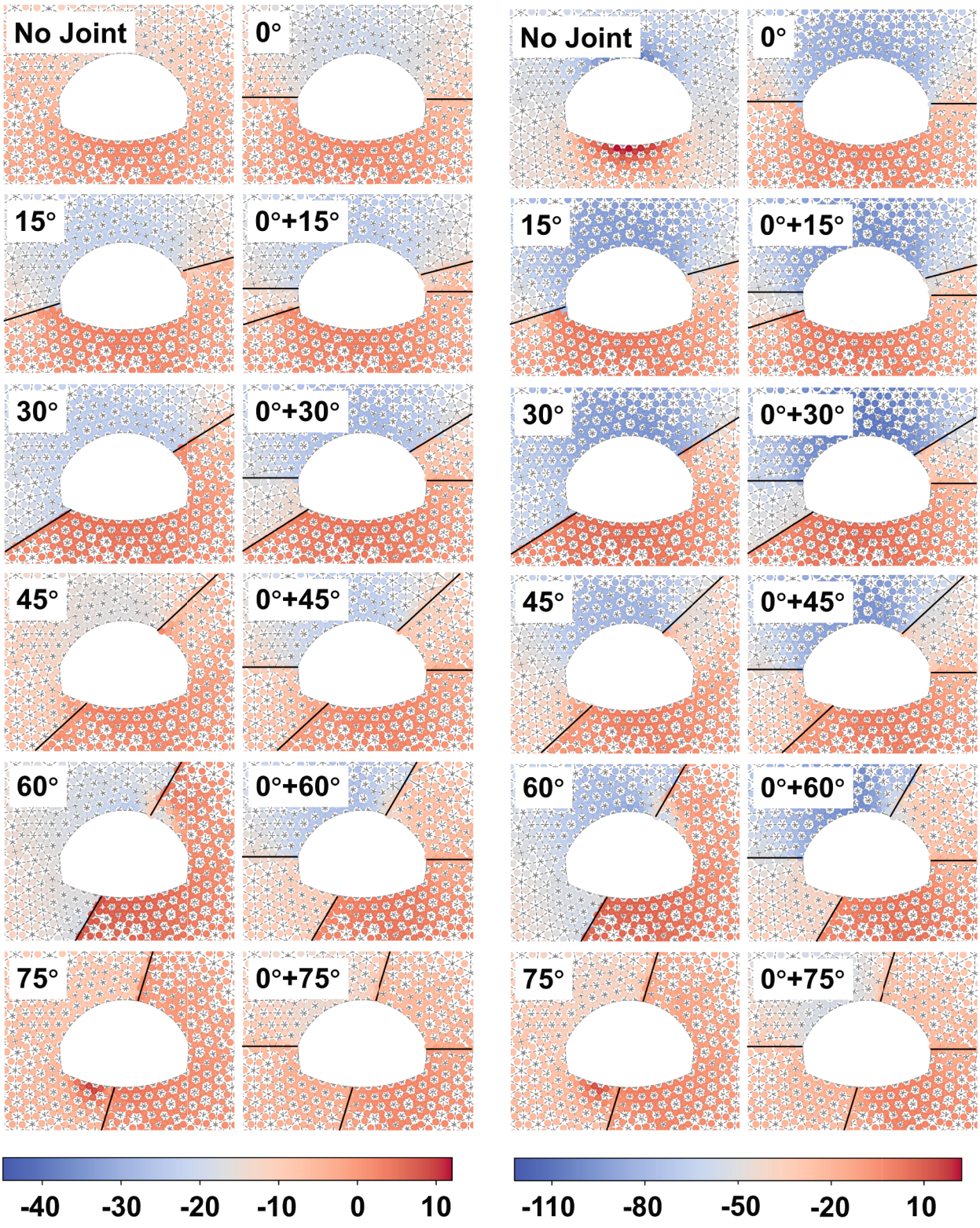


Fig. 12. Vertical displacement nephogram around the tunnel under different lateral pressure coefficients and joint distributions.

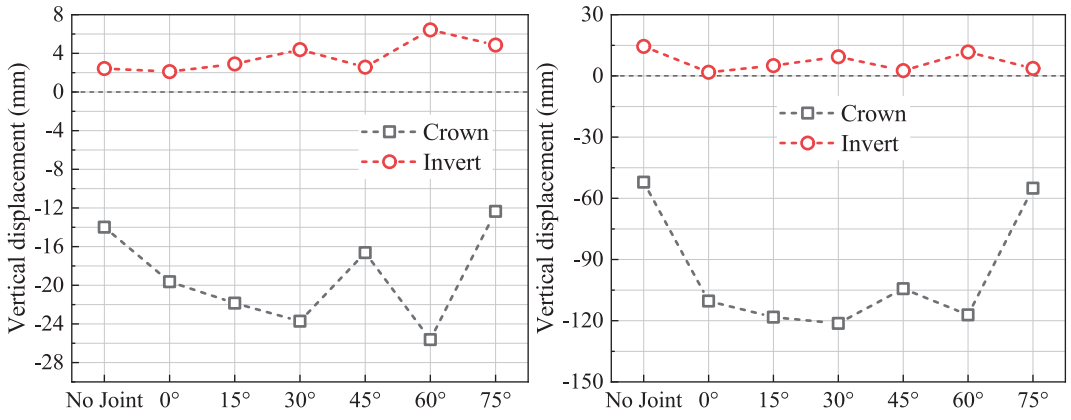


Fig. 13. Vertical displacement of the tunnel under different lateral pressure coefficients and single-set joint distributions.

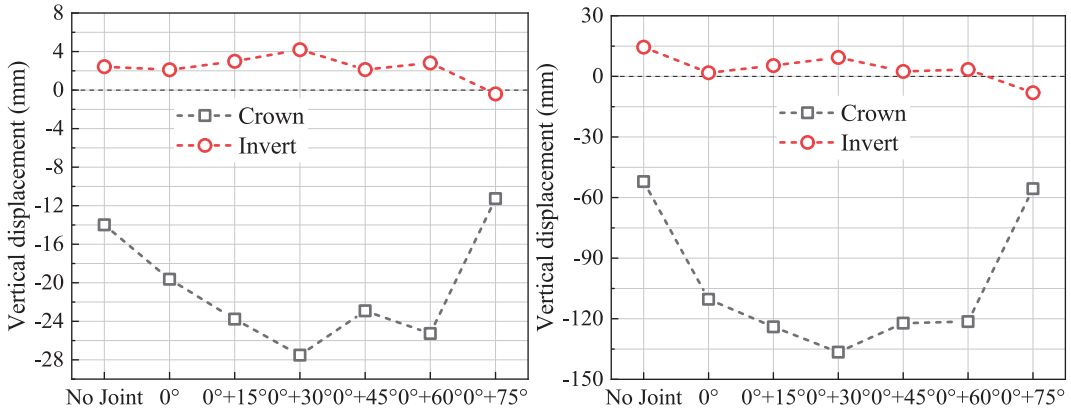


Fig. 14. Vertical displacement of the tunnel under different lateral pressure coefficients and double-set joint distributions.

changes in joint dip angle. This further indicates that in jointed rock masses, the spatial geometric characteristics of structural planes reconstruct the stress transfer path and failure mechanism of the rock mass, dominating the displacement distribution pattern, while the lateral pressure coefficient only affects the absolute magnitude of displacement, and its effect is far weaker than the controlling effect of joints.

5.3. *Influence of GSI*

As an important parameter in the GZZ and GHB strength criteria, *GSI* represents the degree of rock mass fragmentation and reflects the

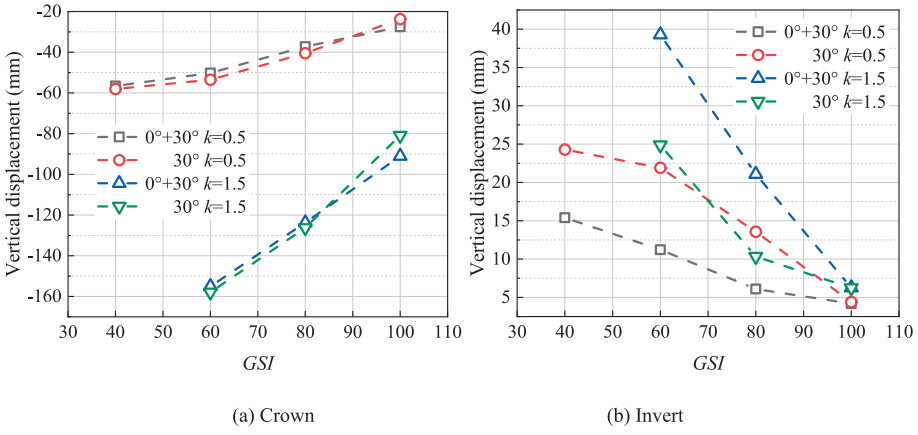


Fig. 15. Influence of different *GSI* values on the vertical displacement of the tunnel.

discontinuity of the rock mass to a certain extent. Therefore, the *GSI* value was varied while considering the presence of joints to investigate the variation trend of tunnel displacement. This section explores the joint effects incorporated simultaneously at macroscopic engineering scale and microscopic intact-rock scale. The vertical displacements of the tunnel crown and invert are shown in Fig. 15, and the plastic zone distribution is extracted in Fig. 16.

When the lateral pressure coefficient is large $k = 1.5$, numerical non-convergence occurs when *GSI* is small, indicating that the rock mass strength is excessively low at this time. As the *GSI* value increases from 40 to 100, both the downward displacement of the crown and the upward displacement of the invert show a decreasing trend, reflecting the effect of improved rock mass integrity on deformation control and inhibiting the stress redistribution caused by excavation. Regarding the influence of the lateral pressure coefficient: under the same *GSI*, the downward displacement of the crown when $k = 1.5$ is generally greater than that when $k = 0.5$, revealing that high lateral pressure exacerbates the deformation of the tunnel crown and invert.

However, when $GSI > 80$, the displacement curves of all cases tend to converge, indicating that rock mass integrity has dominated the deformation behavior, and the effect of structural planes is relatively weakened. Therefore, in jointed rock masses, *GSI* quantifies the weakening degree of rock mass integrity by the structural plane network, and its influence weight is far greater than the individual changes in lateral pressure coefficient and joints.

In practical tunnel engineering, parameters including *GSI* within the GZZ constitutive model can characterize rock fragmentation at the mineral grain scale, whereas large-scale discontinuities such as faults at the tunnel engineering scale can be simulated using the continuous-discontinuous calculation of E-NMM, thereby fully reflecting the development status of rock joints in tunnels.

Consistent with the above conclusions, the plastic zone distribution gradually decreases with the increase of *GSI* value, indicating the regulatory effect of *GSI* on rock mass strength.

6. Conclusions

With the core objective of analyzing the stability of deeply buried jointed tunnels, this paper introduces the smoothed GZZ strength criterion into the continuous-discontinuous computational framework of the E-NMM, and systematically reveals the stability control mechanism of deeply buried jointed tunnels. The main research conclusions are as follows:

(1) The numerical integration of the smoothed GZZ constitutive model into E-NMM has been successfully achieved. By introducing the relaxation iteration method, the solution stability of the double nonlinearity problem is improved. The reliability and applicability of the proposed method are verified through numerical validation.
(2) Joint distribution exerts a dominant controlling effect on the stability of deeply buried tunnels. The influence of joints on the distribution of the plastic zone and the morphology of the displacement field is far greater than that of the ground stress level. Under the condition of a single joint, the plastic zone extends directionally along the joint; when two joints overlap, the plastic zone concentrates in the rock bridge area when the angle between the two joints is small, and returns to the single-joint control mode as the angle increases.
(3) The lateral pressure coefficient and *GSI* value affect tunnel deformation and strength characteristics through different mechanisms. The lateral pressure coefficient mainly regulates the absolute magnitude of displacement. As a key parameter characterizing rock mass integrity, an increase in *GSI* can significantly improve rock mass strength, inhibit the expansion of the plastic zone, and reduce displacement development.

However, the numerical model in this paper is limited to the two-dimensional plane strain assumption and neglects uncertain deformation along the longitudinal tunnel axis. In actual deeply buried tunnel engineering, the mechanical response and failure mode of rock masses are essentially three-dimensional problems. Establishing a three-dimensional E-NMM model is the main goal of future research. Meanwhile, this paper only considers the simplified scenario of one or two joints, and the current simplified model is difficult to fully reflect the action mechanism of complex joint networks. The present study fails to fully account for the effect of excavation unloading paths on the mechanical behavior of surrounding rock, which needs further

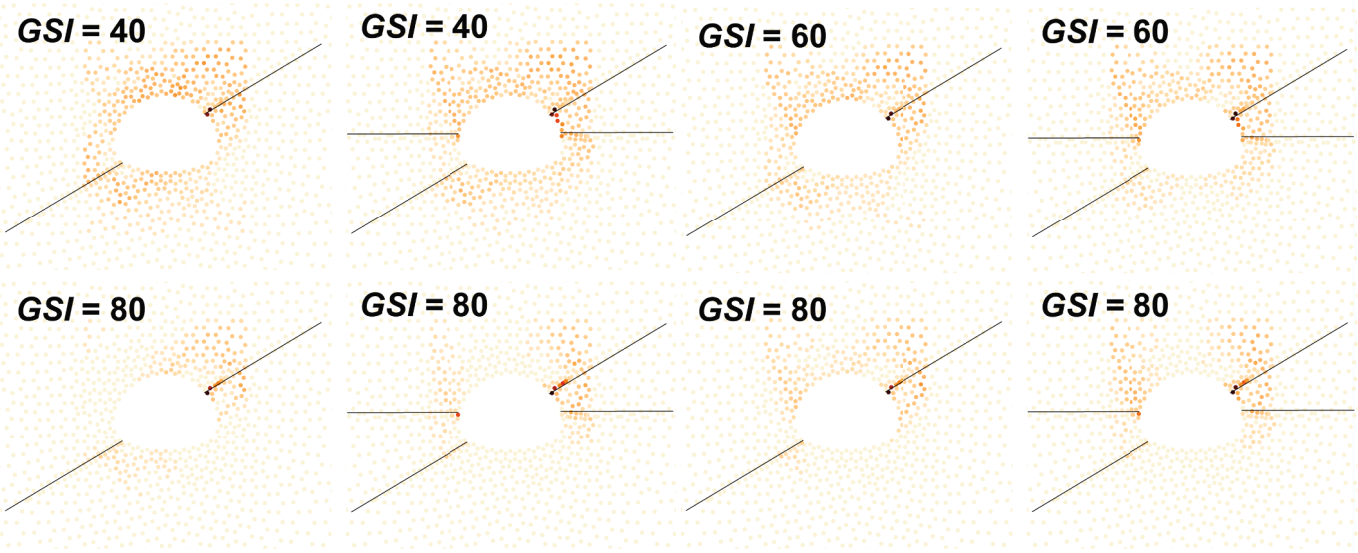


Fig. 16. Plastic zone distribution of the tunnel under different lateral pressure coefficients and *GSI* values.

investigation in subsequent research.

Notably, supporting systems are essential components of tunnel construction. No detailed simulation or experimental validation of supporting structures is carried out herein, and only unsupported tunnel cases are calculated. In engineering practice, jointed tunnels are dominated by asymmetric deformation. Therefore, researches on support design bear great practical significance and will serve as the core content of our future work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Data availability

Data will be made available on request.

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